

ChatGPT:

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</> Python 📄 ▶ Executar

from sympy import symbols, Eq, solve

# 1 - vetores
a = symbols('a')

u = (2, 1, a)
v = (a + 2, -5, 2)
w = (2*a, 8, a)

# 2 - u + v
u_v = (u[0] + v[0], u[1] + v[1], u[2] + v[2])

# 3 - w + u
w_u = (w[0] + u[0], w[1] + u[1], w[2] + u[2])

# 4 - produto escalar
prod = sum(u_v[i] * w_u[i] for i in range(3))

# 5 - ortogonalidade
eq = Eq(prod, 0)

# 6 - resolver
sol = solve(eq, a)

sol
```

Resultado:

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... [-7/4 + sqrt(161)/4, -sqrt(161)/4 - 7/4]
```